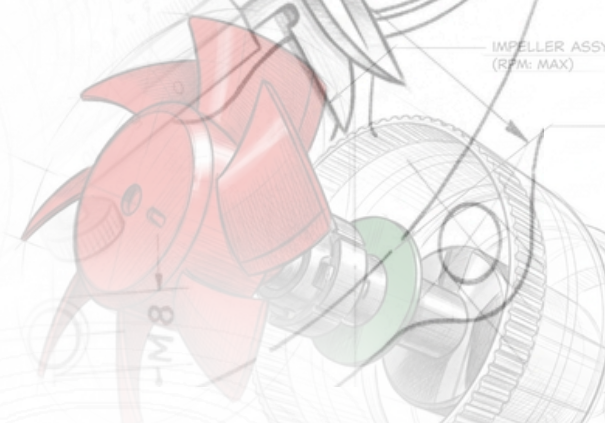
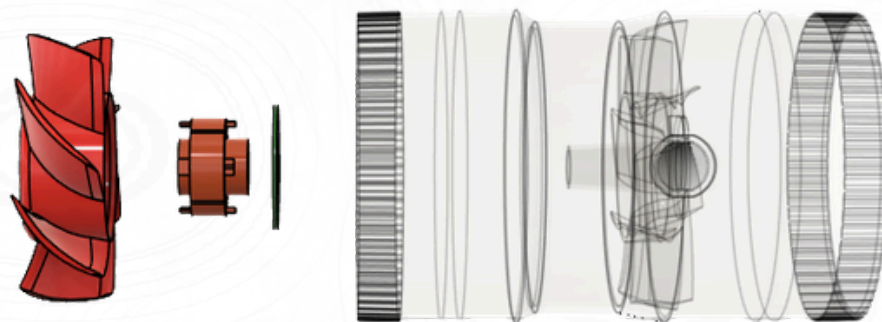


CASE STUDY

{ REVERSE ENGINEERING }
YACHT EXHAUST FAN

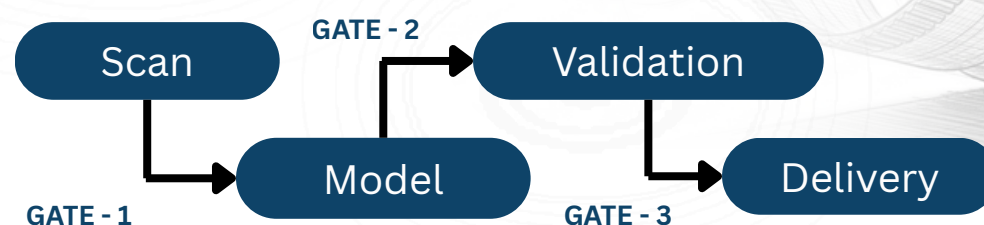
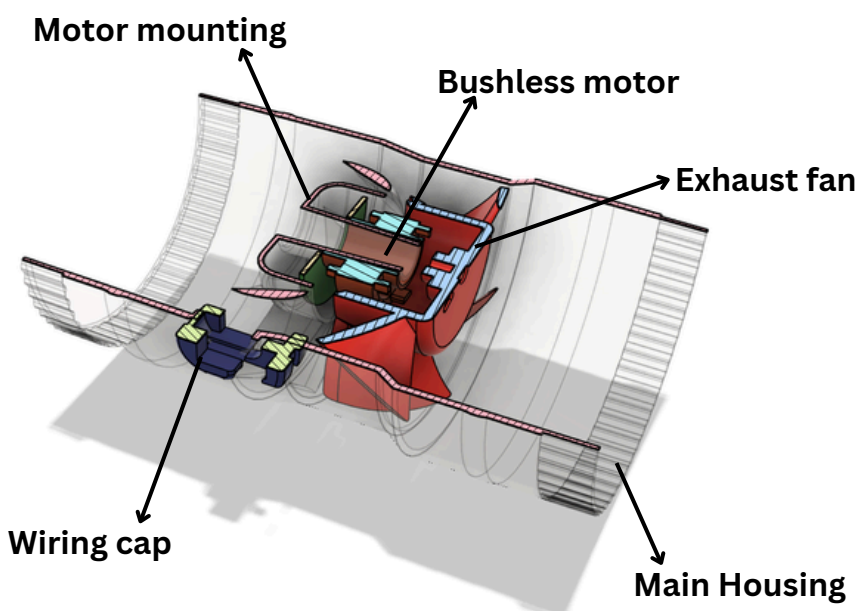


OVERVIEW

The client required detailed design of yacht exhaust fan of a benchmark product but lacked access to its original data, drawings, and specifications. This project focused on reverse engineering a plastic exhaust fan by applying high-precision 3D scanning and CAD modeling to accurately replicate its geometry while achieving functional features.

CHALLENGES

- Non-possibility of exhaust fan disassembly
- Complex blade geometry scanning
- Capturing internal features of exhaust duct
- Estimation and validation of motor mounting unit

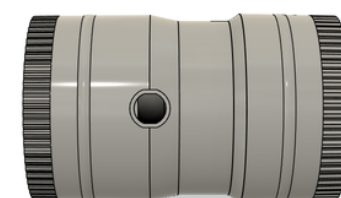


SOLUTION

- Gate - 1 : Due to the complexity of the fan profile and airflow within the duct, it was essential to ensure that scanning process achieves required accuracy levels. The accuracy of the scanning process was validated for both Component 1 and Component 2 as per our SOPs.
- Gate - 2 : This step ensures that our industrial designers follow refined standard operating procedures (SOPs) while modeling the required components such as the fan and duct. Both surface and solid modeling techniques were utilized to develop detailed design assemblies.
- Gate - 3 : This final step verified model (vs) scan comparison to display any discrepancies. It ensures that we achieve the desired results for reverse engineering within acceptable boundary limits.

OUTCOMES

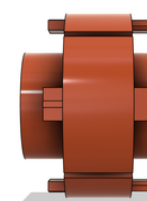
- Project Timeline: 2 weeks
- Digital 3D model delivered
- Cost-effective solution



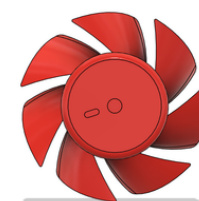
A. Outer Housing



B. Washer



C. Motor

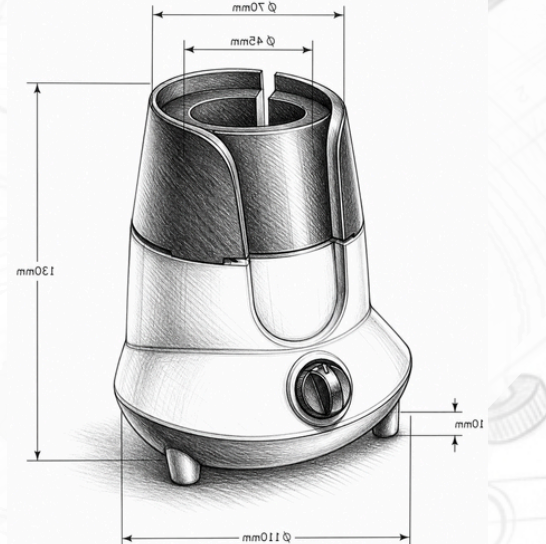
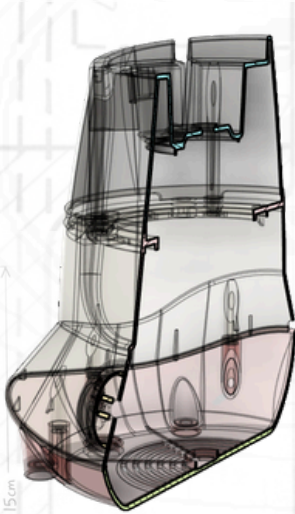


D. Fan

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CASE STUDY

{ REVERSE ENGINEERING } HOUSEHOLD MIXER GRINDER

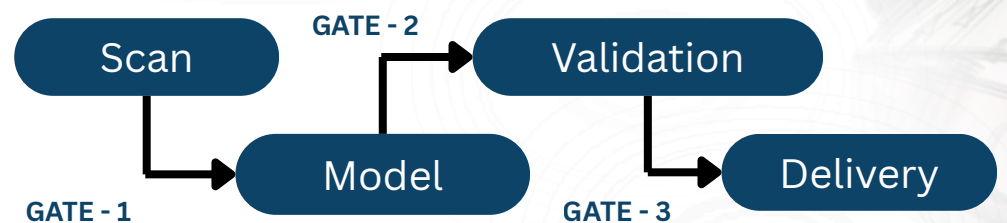
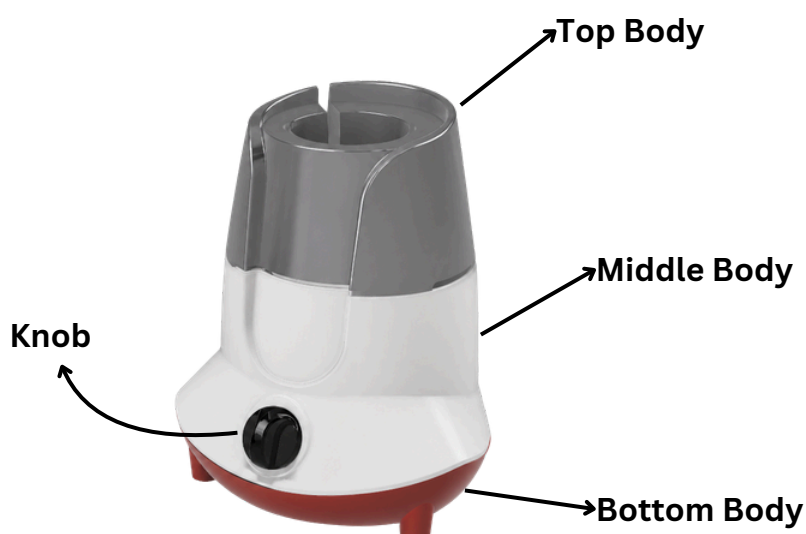


OVERVIEW

This case study focuses on the reverse engineering of a household mixer grinder to understand and replicate its design and functionality. Our client required engineering model of a benchmark mixer grinder to obtain accurate data for design modification, as no original CAD data was available. We provided a 3D CAD data for design analysis and product development.

CHALLENGES

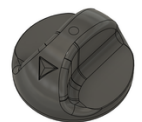
- Capturing grinder complex geometry, functional details and aesthetic features.
- Reflective plastic surface posed a challenge while scanning.
- Adapting surface modeling techniques to accurately mimic scans was the greatest challenge.



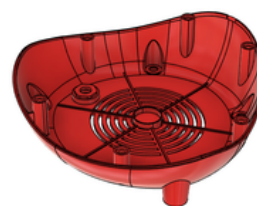
- Gate - 1 : Due to the complexity and aesthetic features of the mixer grinder outer body, it was essential to ensure that the scanning process achieves the desired accuracy levels. The accuracy of the scanning process was validated using engineering trackings.
- Gate - 2 : This step ensures that our designers follow standard operating procedures (SOPs) while modeling assembly components. We utilized surface solid modeling traditional techniques along with tracking digrams to develop detailed design assemblies.
- Gate - 3 : This final step validates our cad design through a model vs scan comparison to identify any discrepancies. It ensures that the reverse engineering process achieves the desired results within acceptable engineering tolerance limits.

OUTCOMES

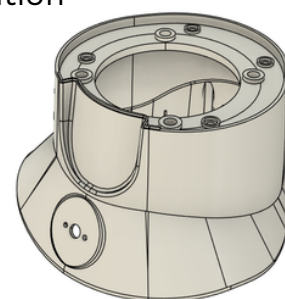
- Project Timeline: 4 weeks
- Digital 3D model delivered
- Cost effective solution



A. Knob



B. Bottom Body



C. Middle Body



D. Top Body



HOUSEHOLD MIXER GRINDER